

# Effective Action and Renormalization in $\phi^3$ Theory

Source: Perimeter Scholars International, *Quantum Field Theory II*, Tutorial 5; 7 worked parts.

## CONTENTS

- 1 The Effective Action for  $\phi^3$  Theory
- 2 Renormalization

Question prompts paraphrased from the official Perimeter PSI Online sheet; the equations and notation follow the source.

## 1 The Effective Action for $\phi^3$ Theory

In six-dimensional Euclidean spacetime, consider

$$S_\lambda = \int d^6x \left[ \frac{1}{2}(\partial\phi)^2 + \frac{m^2}{2}\phi^2 + \frac{\lambda}{3!}\phi^3 \right]. \quad (1.1)$$

For a bosonic theory, the one-loop effective action is

$$\Gamma[\phi] = S[\phi] + \frac{\hbar}{2} \text{Tr}[\ln(S''[\phi])] + O(\hbar^2), \quad (1.2)$$

where terms of order  $\hbar^n$  correspond to  $n$ -loop diagrams.

1. Calculate the integral kernel associated with the operator  $S''[\phi]$  in  $\phi^3$  theory.

**Solution.**

$$S''[\phi]_{xy} = (-\partial^2 + m^2 + \lambda\phi(x))\delta^d(x-y). \quad (1.3)$$

2. Factor the free-field contribution out of  $S''[\phi]$  and expand  $\text{Tr}[\ln(S''[\phi])]$  perturbatively in  $\lambda$ .

*Hint:* Recall

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} - \frac{x^6}{6} + \dots \quad (1.4)$$

**Solution.** Write

$$(-\partial^2 + m^2)\Delta(x-y) = \delta^d(x-y) \iff (-\partial^2 + m^2)_{xy}\Delta_{yz} = \delta_{xz}. \quad (1.5)$$

Then, we have

$$S''[\phi]_{xy} = (-\partial^2 + m^2)_{xz} (\delta_{zy} + \Delta_{zy} \lambda \phi(z)). \quad (1.6)$$

This immediately follows

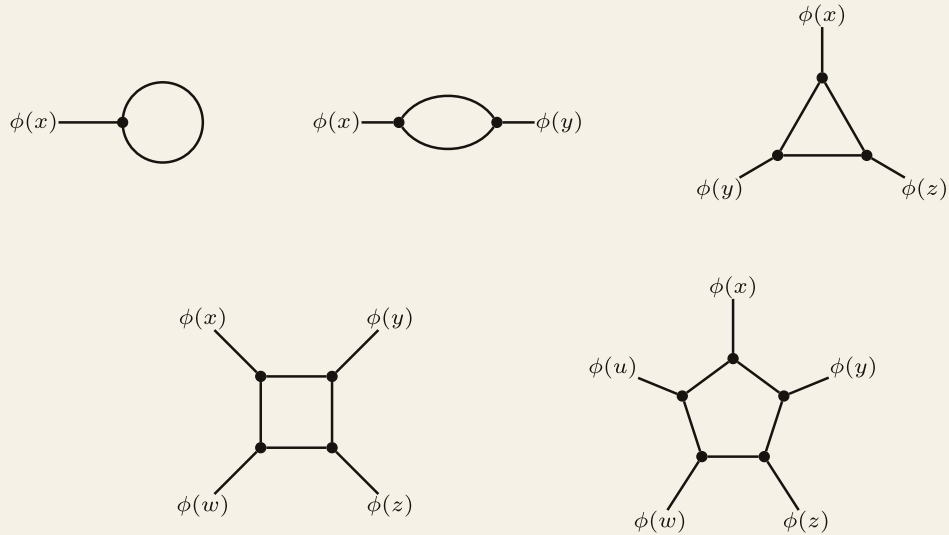
$$\begin{aligned} \text{Tr} \ln(S''[\phi]) &= \text{Tr} (\ln(-\partial^2 + m^2)_{xz} + \ln(\delta_{zy} + \lambda \Delta_{zy} \phi(z))) \\ &= \ln(\det(-\partial^2 + m^2)) + \text{Tr} \left( \lambda \Delta_{zy} \phi(y) - \frac{\lambda^2}{2} \Delta_{zu} \Delta_{uy} \phi(z) \phi(u) + \dots \right) \\ &= \ln(\det(-\partial^2 + m^2)) + \lambda \Delta(0) \int d^d x \phi(x) - \frac{\lambda^2}{2} \int d^d x d^d y \phi(x) \Delta^2(x-y) \phi(y) + \dots \end{aligned} \quad (1.7)$$

3. Express the complete one-loop effective action  $\Gamma[\phi]$  as a power series in  $\lambda$  through fifth order. Represent the result using one-loop amputated Feynman diagrams.

**Solution.** Up to fifth order in  $\lambda$ , we have

$$\begin{aligned} \Gamma[\phi] &= \int d^d x \left[ \frac{1}{2} (\partial \phi)^2 + \frac{m^2}{2} \phi^2 + \frac{\lambda}{3!} \phi^3 + \frac{\hbar \lambda}{2} \Delta(0) \phi(x) \right] \\ &\quad - \frac{\hbar \lambda^2}{4} \int d^d x d^d y \phi(x) \Delta^2(x-y) \phi(y) \\ &\quad + \frac{\hbar \lambda^3}{6} \int d^d x d^d y d^d z \Delta(x-y) \Delta(y-z) \Delta(z-x) \phi(x) \phi(y) \phi(z) \\ &\quad - \frac{\hbar \lambda^4}{8} \int d^d x d^d y d^d z d^d w \Delta(x-y) \Delta(y-z) \Delta(z-w) \Delta(w-x) \phi(x) \phi(y) \phi(z) \phi(w) \\ &\quad + \frac{\hbar \lambda^5}{10} \int d^d x d^d y d^d z d^d w d^d u \Delta(x-y) \Delta(y-z) \Delta(z-w) \Delta(w-u) \Delta(u-x) \\ &\quad \times \phi(x) \phi(y) \phi(z) \phi(w) \phi(u) \\ &\quad + \mathcal{O}(\lambda^6) + \mathcal{O}(\hbar^2). \end{aligned} \quad (1.8)$$

where we have ignore overall constants. These corresponds to the diagrams



4. Explain how the coefficients in the expansion from the preceding part are related to the symmetries of the corresponding graphs.
5. Use the one-loop effective action to show that quantum effects in  $\phi^3$  theory generate a vacuum expectation value for  $\phi(x)$ .

*Hint:* The quantum vacuum field  $\varphi(x)$  is the configuration that extremizes the quantum effective action.

**Solution.** Since there is now a term of order 1 in  $\phi$ , there is a constant term in the effective equation of motion,  $\Gamma'[\varphi] = 0$ . Therefore,  $\varphi$  must be nonzero, and by the definition of  $\varphi$ , the VEV of  $\phi^3$  theory is nonzero.

## 2 Renormalization

1. Redefine the field so that it creates a normalized one-particle state with no vacuum overlap:

$$\langle \Omega | \phi(0) | \Omega \rangle = 0, \quad \langle \mathbf{k} | \phi(0) | \Omega \rangle = 1. \quad (2.1)$$

Show that the Lagrangian can then be written as

$$\mathcal{L} = \frac{1}{2} A \partial_\mu \phi \partial^\mu \phi + B \phi + \frac{1}{2} C \phi^2 + \frac{1}{3!} D \phi^3. \quad (2.2)$$

**Solution.** In general, the bare field would give

$$\langle \Omega | \phi_0(0) | \Omega \rangle = v, \quad \langle \mathbf{k} | \phi_0(0) | \Omega \rangle = \sqrt{Z}. \quad (2.3)$$

Define the renormalised field as

$$\phi(x) = \frac{1}{\sqrt{Z}} \phi_0(x) - v, \quad (2.4)$$

so that

$$\langle \Omega | \phi(0) | \Omega \rangle = 0, \quad \langle \mathbf{k} | \phi(0) | \Omega \rangle = 1. \quad (2.5)$$

Plugging this into the Lagrangian, we find

$$\mathcal{L} = \frac{1}{2} A \partial_\mu \phi \partial^\mu \phi + B \phi + \frac{1}{2} C \phi^2 + \frac{1}{3!} D \phi^3 \quad (2.6)$$

after some redefinition.

2. The factor  $A$  ensures the equal-time canonical commutator  $[\phi(\mathbf{x}), \Pi(\mathbf{y})] = i\hbar\delta(\mathbf{x} - \mathbf{y})$ . Use the six-dimensional Källén--Lehmann representation

$$\begin{aligned} \langle \Omega | \phi(x) \phi(y) | \Omega \rangle &= \int \frac{d^5 \mathbf{k}}{(2\pi)^5} \frac{e^{ik \cdot (x-y)}}{2E_{\mathbf{k}}} |\langle \mathbf{k} | \phi(0) | \Omega \rangle|^2 \\ &\quad + \int_{4m^2}^{\infty} ds \int \frac{d^5 \mathbf{k}}{(2\pi)^5} \frac{e^{ik \cdot (x-y)}}{2E_{\mathbf{k}}} \rho(s) \end{aligned} \quad (2.7)$$

to find the relation between  $A$  and  $\rho(s)$ .

**Solution.** Consider

$$\begin{aligned}
\partial_{y_0} \langle \Omega | [\phi(x), \phi(y)] | \Omega \rangle \Big|_{x_0=y_0} &= \frac{i}{2} \int \frac{d^5 \mathbf{k}}{(2\pi)^5} (e^{-i\mathbf{k} \cdot (\mathbf{x}-\mathbf{y})} + e^{i\mathbf{k} \cdot (\mathbf{x}-\mathbf{y})}) \\
&\quad + \frac{i}{2} \int_{4m^2}^{\infty} ds \int \frac{d^5 \mathbf{k}}{(2\pi)^5} (e^{-i\mathbf{k} \cdot (\mathbf{x}-\mathbf{y})} + e^{i\mathbf{k} \cdot (\mathbf{x}-\mathbf{y})}) \rho(s) \\
&= i\delta^5(\mathbf{x}-\mathbf{y}) + i \int_{4m^2}^{\infty} ds \rho(s) \delta^5(\mathbf{x}-\mathbf{y}).
\end{aligned} \tag{2.8}$$

Meanwhile, we have

$$\Pi(x) = Z \partial_0 \phi(x). \tag{2.9}$$

Therefore, we have

$$\partial_{y_0} \langle \Omega | [\phi(x), \phi(y)] | \Omega \rangle \Big|_{x_0=y_0} = \frac{1}{Z} \partial_{y_0} \langle \Omega | [\phi(\mathbf{x}), \Pi(\mathbf{y})] | \Omega \rangle \implies 1 = Z \left( 1 + \int_{4m^2}^{\infty} ds \rho(s) \right). \tag{2.10}$$

3. Impose four renormalization conditions to determine  $A, B, C, D$ . Two are the normalization conditions above; a third sets the physical mass to zero. For the fourth, require

$$\Gamma(p_1, p_2, p_3) = \lambda_R \quad \text{when} \quad (p_1 + p_2)^2 = (p_1 + p_3)^2 = \mu^2. \tag{2.11}$$

With a sharp momentum cutoff  $\Lambda$ , determine  $A, B, C$  in terms of  $\lambda_R$  and  $\Lambda$  to the order requested, and determine  $D$  through order  $\lambda_R^2$ .

**Solution.** First, let us rewrite the Lagrangian as

$$\mathcal{L} = \frac{1}{2}(\partial\phi)^2 + \frac{1}{2}m^2\phi^2 + \frac{\lambda}{3!}\phi^3 + \frac{\delta_A}{2}(\partial\phi)^2 + B\phi + \frac{1}{2}\delta_C\phi^2 + \frac{\delta_D}{3!}\phi^3, \tag{2.12}$$

where  $\delta_A = A - 1$ ,  $\delta_C = C - m^2$ , and  $\delta_D = D - \lambda$ . Let us start with  $B$ , at one loop order, the only contributing diagram is

$$\text{---} \bullet \text{---} \bigcirc \text{---} + \text{---} \times \text{---} = -\frac{\lambda}{2} \int \frac{d^6 k}{(2\pi)^6} \frac{1}{k^2 + m^2} - B. \tag{2.13}$$

Since the vacuum expectation value must vanish, the above must vanish, and hence we have

$$B = -\frac{\lambda}{128\pi^3} \left[ \frac{\Lambda^4}{4} - \frac{m^2 \Lambda^2}{2} + \frac{m^4}{2} \ln \left( 1 + \frac{\Lambda^2}{m^2} \right) \right] \tag{2.14}$$

at one-loop order.

Now, we turn to the mass renormalisation factor. At 1-loop order, we have

$$\begin{aligned}
-M_2(p^2) &= \text{---} \bullet \text{---} \bigcirc \text{---} \bullet \text{---} + \text{---} \times \text{---} \\
&= \frac{(-\lambda)^2}{2} \int \frac{d^6 k}{(2\pi)^6} \frac{1}{(k^2 + m^2)((k+p)^2 + m^2)} - \delta_C - \delta_A p^2.
\end{aligned} \tag{2.15}$$

The renormalisation condition requires

$$m^2 + M_2(0) = 0, \quad \lim_{p^2 \rightarrow 0} \frac{p^2}{p^2 + m^2 + M_2(p^2)} = 1 \implies M_2(0) = -m^2, \quad M_2'(0) = 0. \quad (2.16)$$

Therefore, we have

$$C = \frac{\lambda^2}{2} I_2(0) = \frac{\lambda^2}{256\pi^3} \left[ \Lambda^2 + \frac{m^2 \Lambda^2}{\Lambda^2 + m^2} - 2m^2 \ln \left( 1 + \frac{\Lambda^2}{m^2} \right) \right], \quad (2.17)$$

$$A = 1 + \frac{\lambda^2}{2} I'(0) = 1 - \frac{\lambda^2}{768\pi^3} \left[ \ln \left( 1 + \frac{\Lambda^2}{m^2} \right) + \frac{2m^2}{\Lambda^2 + m^2} - \frac{m^4}{2(\Lambda^2 + m^2)^2} - \frac{3}{2} \right]. \quad (2.18)$$

Since the first loop correction to the three-point function is of order  $\lambda^3$ . At order  $\lambda^2$ , we merely have  $D = \lambda_R$ .